

Stage 1 — The Policy Environment

**Review the policy env's status quo, implement the
reform**

Where You Are



The Policy Environment

GETTSIM's **policy environment** collects

- policy functions,
- parameters, and
- parameter functions

for a given **policy date**.

It is fully customizable: you can change any parameter or function, and see what happens.

```
{
  ...
  "bürgergeld": {
    ...
    "regelbedarf_m": PolicyFunction
    "parameter_anrechnungsfreies_einkommen_ohne_kinder_in_bg":
      PiecewisePolynomialParam
    "p_id_einstandspartner": PolicyInput
  },
  "kindergeld": {
    ...
    "satz_m": ScalarParam
    "betrag_m": PolicyFunction
  },
}
```

Suggested Workflow

1. Look at the implementation of the current law in your local clone of GETTSIM. Find out which parameters you need to change.
2. Create some input data. Pass it to GETTSIM using the status quo environment. Get familiar with how the output looks like.
3. Call `gettsim.main` with the policy environment as target.
4. Create the new functions and parameters. Add them to the policy environment you got in the last step.
5. Use the input data from the second step and pass it to `gettsim.main`, this time with the modified policy environment.

Tip: Ask GETTSIM what input data you need

```
main(  
    main_target=MainTarget.templates.input_data_dtypes.tree,  
    tt_targets=TTTargets.tree(TARGETS),  
    policy_date_str=POLICY_DATE,  
)
```

Task

Until 14:45 — the status quo

1. Open `cp notebooks/01_policy_environment.ipynb`
2. Find your namespace in `sorted(status_quo)` . Print its parameters and functions.
3. Which leaf is your reform? Read it. `type()` it.
4. Use some fake input data to get tax/transfer results.

15:30–16:30 — the reform

5. `copy_environment` , change the leaf, run again.
6. Keep both environments around. Tomorrow picks them up.

Stuck? Ask Hans-Martin or me.